

COMP5434 Project

1. Overview

This project requires teams to develop a solution for predicting review ratings using a provided dataset. The objective is to build a method that is both accurate and computationally efficient, leveraging big data parallel/distributed processing systems.

2. Deadlines

- a. Kaggle Predictions Submission: **11:59 PM, July 1st, 2026**
- b. Presentation slides, report and code submission on blackboard: **11:59 PM, July 1st, 2026**

3. Team Requirements

Students should participate in this project in teams. Each team should have a voluntary coordinator for administrative purposes. The coordinator should fill in a form (PolyU Connect accounts required, <https://forms.office.com/r/3GqN7HjDri>) to register the team on or before **11:59 PM 12th June, 2026**. Students not in a team by this deadline will be assigned randomly.

4. Data Description

For the data, we provide its training data (train.csv), extra product information (prodInfo.csv), testing data (test.csv), and submission sample (sample_submission.csv). You need to download the dataset and upload your prediction results at the specific Kaggle competition:

<https://www.kaggle.com/t/9e897d08dba249bb8a1312666e8ef8fd>

Your goal is to train a model to predict **the rating of each review**.

Training data: *train.csv*

- Each row corresponds to a single review and contains the following fields: *id* (the id of the review), *title* (the review title), *comment* (the review text), *user_id* (ID of the user who posted the review), *prod_id* (ID of the reviewed product), *parent_prod_id* (Parent ID of the product. For example, same products with different colors share identical parent ID), *time* (timestamp when the review was posted), *votes* (number of users who found the review helpful), *purchased* (whether the user purchased the product), and ground-truth *rating* (an integer from 1 to 5).
- The last cell of each row ("*rating*" column) is the prediction target
- Training data includes training records with ground truth.

Extra product information *prodInfo.csv*

- Each row corresponds to a product and contains the following fields: *id* (row number of this product in this file), *parent_prod_id*, *main_category* (the category of the product), *price*, *title* (name of the product), *features* (description of the product), *store* (the store selling the product), *rating_number* (number of ratings received for this product).

You can use train.csv and prodInfo.csv together to train your solution.

Testing data: *test.csv*

- Each row is a review. The attributes are the same as those of *train.csv*, but the *rating* column is missing.
- Use your method to get your predicted ratings of the testing records in *test.csv* and generate submission file which includes your predictions in the format of *sample_submission.csv*.
- We will obtain your solutions' performance based on your prediction for testing data.

Sample Submission: *sample_submission.csv*

- This file is a sample of your submission about the predictions of *test.csv*.
- This file includes two columns.
 - o *id*: the identifier of each review record in *test.csv*.
 - o *rating*: the predicted rating of this review.

5. Requirements

- Develop a solution to predict review ratings that is both accurate and efficient. The solution **must** incorporate at least one big data parallel/distributed processing framework (e.g., Hadoop, Spark, Flink, Storm, or any other framework that you think is necessary). You may use any algorithms or techniques, *whether covered in the course or not*
- Any programming language is permitted. Your code must be clean, well-documented (with sufficient comments), and accompanied by a high-quality README providing step-by-step execution instructions for reproducibility. Your code will be run to verify the reported efficiency metrics required below.

5.1 Effectiveness

- Effectiveness evaluation is based on the RMSE (root mean squared error) metric. Submit your predictions to the Kaggle competition at:
<https://www.kaggle.com/t/9e897d08dba249bb8a1312666e8ef8fd>
- We use Kaggle (<https://kaggle.com>) to evaluate the accuracy of your predictions. You should attend them as a team. **Your team name in Kaggle must be the same as the name in our team list**, which will be released after the registration period.
 - o Public Leaderboard: Displays your RMSE and rank based on a subset of the test data immediately after submission
 - o Private Leaderboard: Revealed after the competition ends, based on the remaining test data. This score will be used to grade your methods.
 - o **Select two submissions before the competition deadline.** Your final score will be the best of these two on the private leaderboard. If no selection is made, Kaggle automatically uses your highest-scoring public leaderboard submission.
- Some Hints:
 - o Approach the problem as either a regression or classification task.
 - o Consider integrating natural language processing (for text fields like title and comment), graph-based methods (to model user-product relationships), or a combination of both.

5.2 Efficiency

- o Your solution must incorporate at least one big data parallel/distributed processing framework, whether covered in the course or not.

- Resource Options:
 - Use a single machine configured in pseudo-distributed mode to simulate parallel processing.
 - Alternatively, use the departmental HPC platform: <https://hpc.comp.polyu.edu.hk/howto/>
 - Any cloud services you think necessary
- **Mandatory Report Specifications:** In your report, please report
 - **Hardware details:** CPU model, number of servers/threads, GPU model (if applicable).
 - **Efficiency evaluation:** Training time per epoch, Total offline time (preprocessing + training), Total inference time for all test records

6. Report File

- Include your team name, your names, your student ids in the report
- *Suggested report structure:*
 - Problem definition
 - Data Analysis
 - Solution and implementation details
 - Probably follow the major steps in the 5-step big data analytical process
 - Your unique technical designs to improve effectiveness and efficiency
 - Your solution architecture and technical details of each component
 - Any other content that highlights your own designs
 - Performance evaluation and discussion
 - Report RMSE and any additional evaluation metrics considered relevant.
 - Include hardware specifications: CPU model, number of servers/threads, GPU model (if used).
 - Report timing metrics: training time per epoch, total offline time (preprocessing + training), and total inference time for all test records.
 - Analyze experimental results, including ablation studies or comparative evaluations.
 - Draw meaningful conclusions from the findings.
 - Summary of discoveries and future work.
 - Contribution table
 - Provide a table indicating each member's percentage contribution (total 100%). If omitted, equal contribution will be assumed, and grades adjusted accordingly.
 - References

7. Presentation Guidelines

- 5-10 slide explaining your solution
- **Face2Face presentation**
- Each team will have about 10 minutes for a presentation [*May change, depending on the number of teams*].

8. Submission Guidelines:

- Submit your prediction results on Kaggle to obtain your performance on the public leaderboard.
- Then submit the following to **Blackboard**:
 - Create folder with name *TeamName*.
 - o Put all your code together in a folder with name *code*.
 - o Rename your slides with name *slides* (with the extension name, such as *pptx* and *pdf*).
 - o Rename your report with name *report* (with the extension name, such as *pdf*).
 - o Put *code*, *slides* and *report* in the folder *TeamName*.
 - o You should replace *TeamName* to your actual team's name.
 - Compress this folder as one zip file.
 - Follow the below example to name your zip file by replacing “*TeamName*” to your actual team's name:
 - o *TeamName.zip*
- Your submission should be submitted by your TEAM COORDINATOR only

9. Grading Rubrics

- Code (6%):
 - a. Code quality
 - i. Technical designs for efficiency and accuracy
 - b. Code performance on testing data
- Report (5%)
- Presentation (4%)

Outcome Presentations	%	A+/A/A-	B+/B/B-	C+/C/C-	D+/D	F
Code	6%	Programs are well-organized, makes good use of whitespace and comments. Variables have helpful names.	Programs are well-organized, easy to read and understand.	Programs can be read and is in a logical order.	Programs are runnable but barely readable.	Absent
Report Content	5%	Excellent, comprehensive and in-depth analysis with concrete facts/evidence	Clear analysis with good analysis supported by plenty of facts/evidence	Basic analysis with some level of facts/evidence	Barely relevant analysis with minimal facts/evidence	Absent
Presentation quality	4%	Very clear and logical	Good, easy to follow	Understandable, structured	Barely understandable	Absent